

Kabadiwala Connect — Complete Product & Feature Specification

Read this fully before writing any code. This is the single source of truth for what we're building — if a screen or feature isn't described here, it's not in scope for the 4–5 day prototype.

0. The One-Paragraph Version

Kabadiwala Connect is a three-tier system connecting the informal e-waste chain to the formal recycling system. A **door-to-door kabadiwala** gets a zero-friction way to check fair prices and safety info before selling. A **scrap dealer** (our actual primary user) logs what they buy from kabadiwalas, pools it into sellable lots, and hands it over to an authorized recycler with full digital proof — which also happens to be the GST paperwork they now legally need. A **recycler** gets documented, EPR-eligible material they can legally count toward compliance targets, and pays a premium for it because that documentation has real financial value to them. The premium flows back down the chain — that's what makes this an economically real system, not just three connected screens.

Compressed timeline note: since we now have 4–5 days, not 7, this spec marks every feature as **[MVP]** or **[Cut if behind]** — build MVP features first, in the order listed, and only touch a cut-if-behind feature once every MVP feature in your area works end-to-end.

1. The Three Users — Why Each One Exists

User	What they are	What they get	Why they'd bother using it
Kabadiwala	Individual door-to-door collector, often low/no literacy, basic or no smartphone	A lite, countless tool: price check + safety info, voice-first	Zero cost, zero setup, zero risk — pure information
Dealer (our primary user)	Has a shop/yard, buys from many kabadiwalas, sells bulk to recyclers	The full app: purchase logging, lot pooling, recycler matching, ledger	Protects their own margins (GST reverse-charge documentation) and gets better recycler rates
Recycler	Authorized/licensed facility needing EPR-compliant material	A dashboard to receive, verify, and confirm pooled lots + auto-generated EPR certificates	Can legally claim EPR credit only for documented material — this is worth real money to them

The core mechanic that ties them together: every transaction gets recorded at one of two levels of formality — a **lightweight purchase log** when a kabadiwala sells to a dealer, and a **full digital handover record** (photo + GPS + timestamp + QR

confirmation) when a dealer sells to a recycler. The dealer is the hinge point between the two — they're the only user who touches both sides.

2. USER JOURNEY 1 — Kabadiwala Lite Tool

Who builds this: whoever's building the Flutter app can implement this as a stripped-down mode within the same app (simplest for a 4-5 day build) — a "Quick Price Check" entry point that skips login entirely.

Screen 1: Language + entry

- No login screen at all. App opens directly to a language picker (Hindi / Marathi / English) shown as flags/text, remembered for next time.
- Big, obvious icon grid of material categories (same 6–7 categories used everywhere else in the app: PCB, CRT, LCD, cable, battery, motor/magnet, mixed plastic).

[MVP] — this screen must exist; it's the entire kabadiwala-facing product.

Screen 2: Price display

- Tapping a category icon shows: today's price for that material, in large text + a 🔊 button that speaks it aloud in the selected language.
- Data comes from the **same Price dataset** the dealer app reads — no separate data source, just a read-only, no-login view into it.

[MVP]

Screen 3: Safety tip (auto-shown or a small icon button)

- One pictorial + audio safety tip relevant to that category (e.g. tapping "battery" shows "don't puncture or burn — handle carefully" with an icon and a voice clip; tapping "PCB" shows "don't use acid to extract metal — sell to a recycler instead").
- Keep this to one static tip per category, pre-recorded/pre-written — don't try to build a dynamic safety-content system in this timeframe.

[MVP]

What does NOT belong here

- No account creation, no transaction history, no photo capture, no weight entry. If a kabadiwala wants to actually create a lot and get paid formally, they've graduated into being effectively a mini-dealer and should use the full dealer flow. Keep this tool exactly as thin as described — resist the urge to add features here.

3. USER JOURNEY 2 — Dealer App (the core product, most of your build effort goes here)

This is a **Flutter mobile app**. The dealer is running a small business, so design for someone standing in a yard full of scrap, checking their phone between physical tasks — not someone sitting at a desk.

Screen 1: Login

- Phone number + OTP. No email, no password.
- **[MVP]**

Screen 2: Home / Dashboard

- Shows: today's date, a running total of "material logged today," a shortcut button to each of the three core actions below, and a small "pending sync" indicator if there's unsynced data.
- [MVP]

Screen 3: Log a Purchase (from a kabadiwala) — *this is your single most important screen*

This is the feature that gives the dealer a **selfish reason** to use the app from day one, independent of anything recycler-related. It's how Hop 1 (kabadiwala → dealer) gets captured.

What it does, step by step:

1. Dealer taps "Log Purchase."
2. Selects material category (icon grid, same categories as everywhere).
3. Enters weight (numeric keypad or slider).
4. Enters price paid (pre-filled with today's market rate from the Price dataset, editable — dealers negotiate, so the actual price may differ from the board rate).
5. Optionally takes a photo (nice-to-have, not blocking).
6. Taps "Save" — this creates a row in the **Material dataset** (`source_type = 'collector-logged'`) tagged to this dealer.
7. **Crucially:** this same entry doubles as the dealer's GST reverse-charge purchase record. Frame this explicitly in the UI — a small label like "This is your compliance record for this purchase" — because that's *why* the dealer cares about filling it in accurately, not because we told them to.

[MVP] — build this before anything recycler-facing. If your team only ships one working feature by demo day, it should be this one

plus the price board.

Screen 4: My Stock / Pending Lots

- A list of all logged-but-not-yet-sold-to-recycler material, grouped by category, showing total accumulated weight per category (e.g. "PCB: 34kg pending").
- This is what lets a dealer see when they've accumulated enough of one material to make a recycler trip worthwhile.
- **[MVP]** – simple, mostly a filtered list view over the Material dataset.

Screen 5: Create a Lot (pool material for sale to a recycler)

1. Dealer selects a material category from their pending stock.
2. Selects which logged purchases to include (or just "select all of category X").
3. App sums the total weight and calculates an estimated sale value using the current recycler-offered rate.
4. This creates a row in a "pooled lot" – conceptually a **Transaction dataset** row in `pending` status, referencing the underlying Material rows.

[MVP] – this is the bridge between Hop 1 and Hop 2. Without it, purchases just pile up with no path to a sale.

Screen 6: Find a Recycler

- Shows nearby authorized recyclers, ranked by: distance, offered rate for this material category, pickup availability, authorization status (pulled from the **Recycler dataset**).
- Dealer picks one and taps "Request Pickup" or "Confirm Sale."
- **[MVP]** – the ranking can be a simple weighted formula (see earlier ML discussion) – no need for anything fancy.

Screen 7: Handover / QR Confirmation

1. When the recycler arrives (or the dealer drops off), the app shows a QR code encoding the lot's unique ID.
2. Recycler scans it on their dashboard (see Journey 3), confirms actual weight, confirms the deal.
3. Once confirmed, this dealer's app updates the lot status to "confirmed," and a **Traceability dataset** row gets created (photo, GPS, timestamp, confirmation reference).

[MVP] – the QR generation itself is simple; the confirmation flow depends on the recycler dashboard being ready, so coordinate timing with whoever's building that.

Screen 8: Earnings Ledger

- A simple list: all confirmed sales, amounts, dates, running total earned this week/month, and – importantly – a running total of what was *spent* buying from kabadiwalas (from Screen 3's logs), so the dealer can see their actual margin.
- **[MVP]** – mostly a read-only view summing existing data, low build cost relative to value shown.

Screen 9: Price Board (same as kabadiwala's, but dealer-facing)

- Current rates per category, with a basic trend indicator (up/down arrow) – this can reuse the simple linear-regression trend logic discussed earlier.
- **[Cut if behind]** – the trend arrow specifically; the raw price display itself should stay MVP since Screen 3 depends on it for pre-filling prices.

Screen 10: EPR/Compliance Export

- A simple "Export my purchase & sale records" button generating a basic PDF or CSV of the dealer's logged purchases and confirmed sales for a date range — this is literally their GST documentation.
 - **[Cut if behind]** — valuable to demo (it's a strong differentiator to show a judge) but not required for the app's core loop to function. Build this after Screens 1–8 all work.
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4. USER JOURNEY 3 — Recycler Dashboard

This is a **web dashboard** (React/Next.js) — recyclers are fixed businesses, a laptop-friendly table-based UI serves them better than a phone app.

Screen 1: Login

- Phone OTP or simple email/password — doesn't need to be fancy.
- **[MVP]**

Screen 2: Incoming Lots Feed

- A table/list of pooled lots from nearby dealers matching this recycler's accepted materials — shows dealer name, material category, weight, distance, estimated value.
- Filterable by material category and status (pending / confirmed).
- **[MVP]** — this is the recycler's main working screen, build it early.

Screen 3: Confirm Handover

1. Recycler scans (or manually enters) the QR/lot code shown on the dealer's app.

2. Sees the dealer's declared photo/category/weight.
3. Enters the actual verified weight from their own scale.
4. Taps "Confirm" — this finalizes the transaction, writes the confirmed Transaction + Traceability records, and triggers the certificate generation (Screen 4).

[MVP] — this is the other half of the QR handover loop from the dealer's Screen 7; build these two screens together, ideally by the same person or in tight coordination, since they're really one feature split across two apps.

Screen 4: EPR Handover Certificate

- Auto-generated PDF the moment a handover is confirmed: dealer name, material category, verified weight, GPS coordinates, timestamp, photo, unique reference number.
- Recycler can download/view this for their own compliance filing.
- **[MVP]** — this is your single strongest "wow" artifact for the judges; a real, downloadable PDF with real data on it lands much better than a description of the feature.

Screen 5: Recycler Profile / Rate Management

- Recycler sets which materials they accept, their offered rate per category, pickup availability, service area — this feeds directly into the Recycler dataset that powers the dealer's matching screen.
- **[MVP]** — needs to exist for Screen 2/3 of the Dealer journey to have anything real to match against; build this early even though it's a recycler-facing screen, because the dealer app depends on this data existing.

Screen 6: Transaction History

- A simple filterable list/export of all confirmed handovers — supports the "documented trail" pitch point.
 - **[Cut if behind]** — nice to show, not core to the working loop.
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5. How the Three Journeys Actually Connect (Recap, Concretely)

1. Kabadiwala checks price (Journey 1) → sells physically to a dealer (nothing digital happens at this exact moment).
2. Dealer logs the purchase (Journey 2, Screen 3) → this is Hop 1, informal but documented.
3. Dealer pools multiple purchases into a lot (Journey 2, Screens 4–5) and finds a recycler (Screen 6).
4. Dealer and recycler physically meet; QR handover happens (Journey 2 Screen 7 ↔ Journey 3 Screens 2–3) → this is Hop 2, fully formal.
5. Certificate generated (Journey 3, Screen 4) → dealer sees it reflected in their ledger (Journey 2, Screen 8) and can export it for GST purposes (Journey 2, Screen 10).

One thing worth everyone on the team internalizing: Journeys 2 and 3 are really one continuous feature (the handover) built as two halves in two different codebases. Whoever owns the Flutter QR-generation screen and whoever owns the React QR-confirmation screen need to agree on the exact data format (what's encoded in the QR, what fields get sent to the sync endpoint) before either of them starts coding — this is the single most likely place for an integration bug to eat your limited time.

6. Non-Functional Requirements (Apply to Every Screen Above)

- **Offline-first (Dealer app only):** every write in Journeys 2's Screens 3–5 must save locally first and queue for sync – a dealer standing in a yard with no signal must still be able to log a purchase. The kabadiwala tool and recycler dashboard can assume connectivity (kabadiwala tool's price data is just cached read-only; recycler dashboard is used at a fixed business location).
 - **Vernacular:** Hindi and Marathi minimum, on every user-facing screen in Journeys 1 and 2. The recycler dashboard (Journey 3) can be English/technical since recyclers are the more literate, business-facing end of the chain.
 - **Low-end device target:** Journey 1 and 2 (the mobile app) need to run acceptably on a ₹5–6K Android phone – keep photo sizes compressed, avoid heavy animations.
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7. Quick-Reference Feature Table (For Dividing Work)

Feature	Journey	Priority	Depends on
Language + price display	Kabadiwala	MVP	Price dataset populated
Safety tips	Kabadiwala	MVP	None
Login (OTP)	Dealer	MVP	Backend auth endpoint

Feature	Journey	Priority	Depends on
Log a Purchase	Dealer	MVP	Price dataset
My Stock / Pending Lots	Dealer	MVP	Log a Purchase working
Create a Lot	Dealer	MVP	My Stock working
Find a Recycler	Dealer	MVP	Recycler dataset populated (Journey 3 Screen 5)
QR Handover (dealer side)	Dealer	MVP	Recycler-side confirm screen ready in parallel
Earnings Ledger	Dealer	MVP	Confirmed transactions existing
Price trend arrow	Dealer	Cut if behind	Basic price display working
EPR/GST export	Dealer	Cut if behind	Ledger working
Recycler login	Recycler	MVP	Backend auth
Incoming Lots Feed	Recycler	MVP	Recycler profile + dealer lots existing
Confirm Handover	Recycler	MVP	Dealer QR screen ready in parallel

Feature	Journey	Priority	Depends on
EPR Certificate generation	Recycler	MVP	Confirm Handover working
Recycler Profile/Rates	Recycler	MVP	Needed early – unblocks Dealer matching
Transaction history export	Recycler	Cut if behind	Confirm Handover working

Next step once everyone's read this: assign one or two people per journey (Kabadiwala tool is small enough to be a side-task for whoever finishes their main screens early), and figure out the exact day-by-day order given we only have 4–5 days – happy to write that split next once the team has agreed on who's taking what.